IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received 16 February 2026, accepted 4 March 2026, date of publication 6 March 2026, date of current version 12 March 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3671689

SURVEY

Enabling Intelligence in Digital Twins Through Semantic Enrichment and Multi-Agent Systems: A Systematic Literature Review

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This work was supported by Technical University of Applied Sciences Mannheim.

ABSTRACT Semantic technologies and Multi-Agent Systems (MAS) are promising key technologies for making Digital Twins (DTs) intelligent. Due to siloed architectures and the lack of interoperability, many current implementations remain isolated and are limited in terms of intelligence and proactivity. Semantic enrichment provides a common basis of understanding that fosters interoperability and contextual awareness across system boundaries, while MAS approaches enable decentralized, autonomous decision-making processes. Against this background, a systematic literature review of the current research landscape from 2019 to June 2025 examines the state of research on semantic and agent-based enablers for intelligent DTs, bringing together fragmented insights and identifying remaining research gaps and their potential. The findings show that the manufacturing industry dominates this field, using standardized data sources and agent-based orchestration for dynamic scheduling and adaptation, while other sectors are exploring customized ontologies and simpler agent schemes. However, many of these implementations are still in their early stages, lacking closed-loop control or fully autonomous decision-making. The conclusions point to the future need for common semantic standards, unified evaluation strategies, and integration with other artificial intelligence techniques to drive autonomy and proactivity of DT systems.

INDEX TERMS Digital twins, multi-agent systems, semantic technologies, systematic literature review.

I. INTRODUCTION

Digital Twins (DTs) have become a key driver of digital transformation in industrial value chains by linking physical assets, operational processes, and manifold data within integrated digital architectures [1]. However, monolithic, isolated structures hinder the interoperability and reuse of DT concepts across system boundaries and domains [2]. Strengthening the links between physical entities, processes, and conceptual models is therefore essential to enable end-to-end value creation [3], [4]. Through synchronized, digital representations of physical entities with bidirectional communication and intelligent decision-making, DTs are paving the way in this direction [5]. In order to empower them with the necessary intelligence, DTs need to become active, online, goal-seeking, and anticipatory [6]. The synthesis with artificial intelligence technologies, such as semantic technologies, or Multi-Agent Systems (MAS), is considered to be a key enabler for industrial transitions [7], [8]. At the same time, DT frameworks must support adaptable, modular, and interoperable architectures [9]. Across all application domains, DT implementations are highly heterogeneous and often remain merely descriptive shadows without closed-loop implementation, making comparability and aggregation difficult [10].

In this context, this Systematic Literature Review (SLR) consolidates peer-reviewed research from 2019 to June 2025 on semantic enrichment and MAS integration as complementary enablers of intelligent DTs following a predefined workflow. A protocol-driven extraction is carried

The associate editor coordinating the review of this manuscript and approving it for publication was Lukasz Wisniewski.

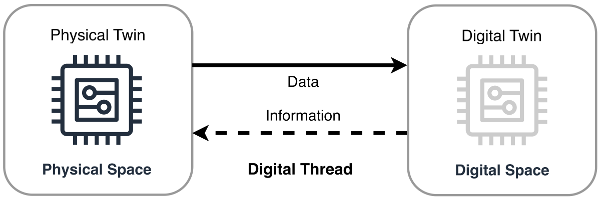

FIGURE 1. Digital twin concept aligned to Grieves [13].

out to synthesize and classify the resulting primary studies. A structured classification based on DT maturity, semantic enrichment, and reported autonomy through agent-based approaches is assessed and further analyzed, which forms a robust basis for comparatively identifying research gaps. This SLR therefore contributes (i) an evidence-based overview of the current research landscape on intelligent DTs at the intersection of semantic technologies and agent systems. (ii) The individual enabling technologies were examined in more detail in various research question clusters and, wherever possible, classified based on existing concepts and taxonomies. (iii) Finally, based on the literature reviewed, research gaps and directions are deduced.

The paper is structured as follows. First, section II outlines related work on DTs, semantic technologies, and MAS, deriving the initial problems addressed by this SLR. Subsequently, the methodology of the literature review is outlined in section III, subdivided into planning, conducting, and reporting phases. The results are assessed in detail in section IV. After section V discusses the results comprehensively, section VI concludes and provides an outlook on future developments.

II. RELATED WORK

The origin and wide range of applications for DTs have evolved from a conceptual paradigm into a widely adopted engineering methodology that addresses common issues, such as optimization, monitoring, control, and real-time prognostics [11]. The constantly growing popularity leads to an increase in further application domains due to its universal applicability [12]. As shown in Fig. 1, the concept pioneered by Grieves emerged from product lifecycle management, in which entities in the physical space correspond bidirectionally with representations in digital space across their lifecycle via the digital thread [13] by analyzing, predicting, and optimizing the real world through respective DTs [14].

Despite the multitude of use cases and domains, DT research and practice still face substantial architectural and terminological heterogeneity [15]. The depth of data integration can be differentiated according to [15] and [16]: A manually created digital representation of a real object without automated data exchange comprises a digital model. The extension of an automatic data flow from real to digital objects encompasses the digital shadow. Bilaterally automated data exchange yields the DT, which can independently reflect adjustments such as optimizations in the real world.

Building on these characteristics, intelligent DTs extend beyond synchronized representations toward autonomous behavior and simulation modeling, leveraging processes to be designed more efficiently and flexibly [17]. This allows intelligent DTs to leverage their advantages proactively, especially in complex systems [18]. In order to provide a tangible understanding of the various evolutionary stages of intelligence in DTs, literature describes the different necessary components as follows: Model understanding creates a semantic understanding of the system entities involved by means of a metamodel or knowledge representation. Intelligent algorithms provide decision logic on this basis, thereby fulfilling a goal-driven optimization process. Asset Services operationalize this and provide executable machine functionalities in an application-oriented manner. The interface closes this control loop between digital and physical space, enabling autonomous behavior to emerge from model comprehension, decision logic, and services [17], [19], [20]. However, intelligence and autonomy are still in an immature stage and must be further pursued in order to meet the premises of Industry 4.0 and 5.0 [7].

A persistent barrier to the implementation of DTs is interoperability, closely linked to the lack of standardization as a key challenge [21], [22]. A common approach to standardization efforts in the field of manufacturing is the Asset Administration Shell (AAS), which is being promoted in particular by the Industry 4.0 Consortium in academia and industry [5], [8]. The objective is to create a universal meta-model with semantic interoperability between manufacturers and applications [3]. There are three distinguished types of AAS, which also correspond to the distinction described above: Passive AAS functions as a static data container for manual exchange (c. f. digital model). Reactive AAS is an online representation with a standardized Application Programming Interface (API) for information exchange (c. f. digital shadow). Proactive AAS expands on this by enabling direct communication and negotiation through distributed decision-making between AAS (c. f. DT) [23]. Other general standardization efforts for DT include ISO 23247 [24], which covers general principles for DT frameworks in manufacturing and standards for the lifecycle and digital continuity (e.g., IEC 62890 [25] and ISO 23704 [26]) address aspects of long-term lifecycle management and continuous operation that are critical during the evolution of industrial DTs. But even different standardizations and their exchange formats are not natively interoperable with each other and require interoperability mappings [27].

Accordingly, semantic technologies aim to enrich DT information semantically with machine-interpretable meaning and context. Therefrom, a common knowledge base and understanding between participating system components can be established via metamodels, ontologies, or knowledge graphs [28], [29]. Heterogeneous data sources can be transposed into homogeneous semantics in order to achieve interoperability through standardized vocabularies that support reasoning and contextualization [30]. Especially ontologies and knowledge graphs are frequently utilized to represent relations and constraints to enhance the functionalities of intelligent DTs [31]. Beyond the semantic requirements, several industrial interoperability and lifecycle data standards are relevant for DTs. In particular, the ISO 10303 [32] family offers a standardized representation and exchange of product data across engineering and manufacturing with integration of product, process, and machining information with the Standard for the Exchange of Product model data (STEP) as shown through an additive manufacturing approach in Rodriguez and Alvares [33]. In plant and production system engineering, AutomationML [34] also provides an open exchange and modeling format for integrating heterogeneous engineering disciplines and tools.

Agent-based approaches such as MAS feature an engineering paradigm for decentralized and parallel decision synthesis leveraging autonomous agents [35]. These are characterized by flexible, intelligent, and autonomous behavior and are designed to fulfill specified objectives while simultaneously adapting to their environment [36]. In Pretel et al. [37], the relationship between DT and MAS is analyzed and differentiated into two main groups: MAS with DT thus function as a service layer for information provision, with agents using this as a basis. MAS for DT, however, take on integral components of DTs and therefore include important core competencies such as simulation, analysis, and decision-making itself. Reported synergy effects between MAS and DTs are assigned to autonomy, sociality, interaction, and proactivity in particular, often relying on a knowledge component for consistent orchestration and negotiation [37].

In general, the literature on intelligent DTs addresses interoperability through semantic enrichment as a prerequisite for distributed intelligence through MAS mechanisms. Nevertheless, the main challenges and problems arise from the fragmented research landscape with different terminology across domains with regard to the combination of enabler technologies and their maturity level [38]. This merging of approaches enables autonomous end-to-end control loops, hence it is important to examine integration and evaluation patterns in current research. Many of the proclaimed DT solutions in practice are only digital shadows, so that changes in the virtual model do not automatically lead to changes in the physical asset, which is also considered to be problematic [37], [39]. Therefore, this SLR consolidates and classifies current findings on approaches to implementing intelligent DTs, with a focus on infrastructure approaches with semantic technologies and MAS integration patterns.

TABLE 1. PICOC criteria.

Population	Intelligent Digital Twins
Intervention	Semantic Enrichment & Multi-Agent Systems
Comparison	Not Applicable
Outcome	Report and meta-analysis on the Current State of the Art and Challenges in semantic and agent-based approaches for intelligent Digital Twins
Context	Peer-Reviewed Publications

III. METHODOLOGY-SYSTEMATIC LITERATURE REVIEW

This section covers the methodological procedure of the SLR, focusing on semantic enrichment and MAS as intelligence-enabling approaches for DTs. In order to establish the methodical framework of this review, the procedure according to Kitchenham and Charters [40] is adapted. The overall process is divided into 3 fundamental steps, which are the planning phase, conducting phase, and reporting phase, as shown in Fig. 2.

Starting with the planning phase, the previously identified and analyzed problem within related work is applied to synthesize the research questions. These serve as the foundation for the subsequent conduction of the SLR. The phase is concluded with the specification of the search mechanisms and the development of a review protocol. During the second phase, primary literature is identified through systematic search strategies, subsequently assessed for relevance, and iteratively refined through a rigorous quality evaluation to facilitate data extraction and analysis. The final phase concerns comprehensive reporting and targeted dissemination of the results in a scientifically suitable manner. The various phases, their inherent evaluation loops, and their execution are presented in more detail in the following sections.

A. PLANNING PHASE

The planning phase marks the initial stage of the SLR, in which the problem is addressed through the preliminary review of relevant literature and processed to identify the research questions. These arise from the aforementioned problems and are formulated using the Population-Intervention-Comparison-Outcomes-Context (PICOC) criteria according to Kitchenham and Charters [40]. Table 1 presents these criteria in the context of intelligent DTs by utilizing semantic technologies and MAS. Based on these criteria, the research questions are then derived and categorized into four distinct groups, as shown in Table 2.

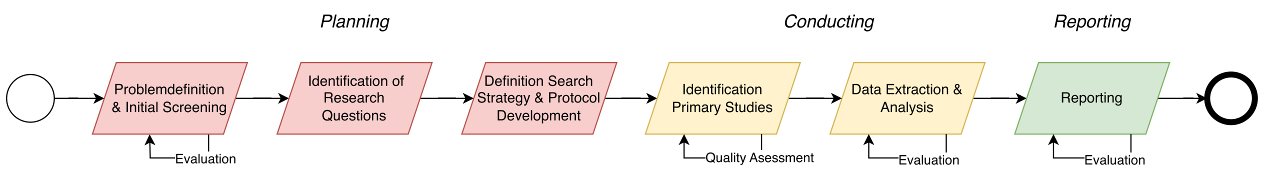

FIGURE 2. SLR methodology aligned to Kitchenham et al. [40].

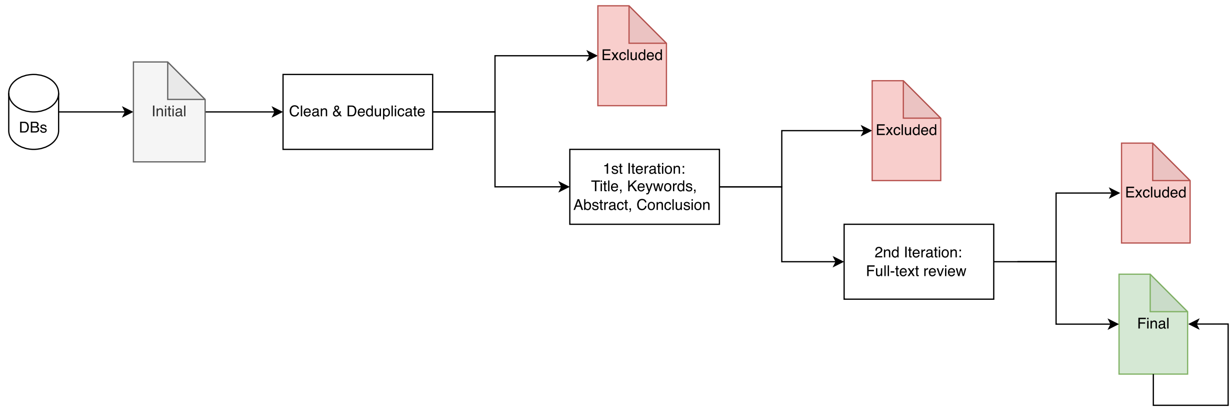

FIGURE 3. Qualitative methodology of the conducting phase.

TABLE 2. Research questions.

RQ 1	Which domains and application scenarios are examined in the study with regard to the combined use of DTs, semantic enrichment of data, and MAS?
RQ 2	Which structures and concepts characterize the DT presented in the study?
RQ 2.1	What level of data integration do the DTs exhibit?
RQ 2.2	To what extent is the standardization approach of the AAS used?
RQ 2.3	How are intelligent aspects taken into account in the DT?
RQ 3	How is the semantic enrichment of the DT accomplished?
RQ 3.1	What conceptual approaches to semantic enrichment are used?
RQ 3.2	Which technologies or standards serve as a framework for semantic enrichment?
RQ 4	What role do MAS play in conjunction with DTs in the study?
RQ 4.1	What level of integration between MAS and DTs does the study indicate?
RQ 4.2	How is the communication between MAS and DT carried out?
RQ 4.3	Which platforms are used to implement the MAS?
RQ 4.4	Which interaction patterns serve negotiation and cooperation between agents and the execution of services on assets?
RQ 5	How are limitations and future work primarily addressed?

After formulating the research questions, an appropriate search strategy must be defined, as illustrated in Fig. 3. This strategy is designed to identify publications that explore the realization of intelligent DTs. The time interval of relevant publications is determined between 2019 and 10.06.2025. The literature search of the databases was carried out using a specific search string. First, all relevant research databases (IEEE, ACM, Science Direct, Dblp, Scopus & Springer Link) are filtered using the following unified search string:

("digital twin" OR "DT" OR "asset administration shell" OR "AAS") AND ("semantic" OR "semantic web" OR "semantic model" OR "semantic enrichment" OR "interoperab*" OR "knowledge graph") AND ("multi-agent*" OR "multi agent*" OR "multiagent*" OR "agent based" OR "agent*" OR "cooperat*" OR "negotiat*"))*

The search string is also derived on the basis of the PICOC criteria from Table 1 and is utilized in an adapted form for the search in the corresponding literature databases. As shown in Fig. 3, the results are subsequently screened for duplicates and refined. Upon completion of the consolidation and cleansing process, the remaining entries form the initial dataset for the first iteration. Predefined inclusion and exclusion criteria are applied to manually examine the papers concerning their title, corresponding keywords, abstract, and, as recommended in [40], the conclusion as well.

TABLE 3. Inclusion / exclusion criteria.

#	Criteria
IC1	The study deals with DTs
IC2	The study deals with the application of MAS
IC3	The study includes or presupposes approaches to semantic enrichment
IC4	The study was published in a peer-reviewed journal or conference proceedings
IC5	The study was published from 2019 onwards
EC1	The paper is not a scientific paper (no educational, editorial, tutorial, material, etc.)
EC2	The paper is a secondary study (survey, systematic review, etc.)
EC3	The paper was written in a language other than English

The inclusion and exclusion criteria, as shown in Table 3 are based both on the research questions and on generally accepted characteristics for valid scientific literature (e. g., peer reviewed). Publications are included in the review only if they satisfy these criteria. Otherwise, these are excluded from further consideration. Subsequent to the application of the criteria, the selected results constitute the foundation for the second iteration. The final step is to conduct a full-text review, taking into account the entire study. The full text review of the second iteration, with the analysis of the research questions within the developed protocol, also includes the quality assessment of the publications. The result of these iterations forms the overall selected publication pool. The iteration process shown is part of the identification of primary studies within the conducting phase and is explained further in the following sections.

B. CONDUCTING PHASE

Following the planning phase with the developed search strategy and associated protocol, the conducting phase can be carried out. The aim here is to analyze the full texts, extract

TABLE 4. Quality assessment criteria.

#	Criterion
Q1	Is the context of the study adequately provided?
Q2	Are the methods and technologies used described in a comprehensible manner?
Q3	Is there an outlook for future research?

data, and analyze them according to specific quality criteria based on the protocol.

During the identification process of the primary studies, as previously shown, it is particularly important to pay attention to specific quality criteria as given in Table 4 of the individual studies. The quality criteria serve as an indicator of whether publications could distort the results of the review. Consequently, these are taken into account in the full-text review and systematically applied to the individual primary studies. If a criterion is fully and comprehensively addressed, a positive rating (+1) is given. If it is only partially recognizable, too brief, or unclear, it is given a neutral rating (0). If the information is missing, a negative rating (-1) is given. If a study falls below an average quality rating of 0, it is excluded from the study pool.

In order to extract the data relevant for the review, a table-based extraction chart was developed and applied based on the research questions. It also encompasses multiple levels, including metadata, inclusion and exclusion criteria, as well as quality assessment criteria, ensuring applicability across all necessary iterations. Throughout the iterations and extraction process, the research questions were further refined, and the entire extraction framework was continuously optimized. The document thus forms the basis for the meta-analysis. All analyses were processed and realized using Pandas and Seaborn libraries in Python.

C. REPORTING PHASE

The final phase focuses on reporting the results of the SLR. The findings are systematically processed, and the research questions are further discussed and evaluated. The focus here is on adequate methods of disseminating the review results.

IV. RESULTS

The literature search was carried out in accordance with the methodology and review protocol introduced previously. Fig. 4 quantitatively shows the results of the individual iteration steps to determine the final primary literature. Following the automated search with the aforementioned search string, 516 initial publications were identified for further examination (IEEE: 123, ACM: 3, Science Direct: 24, Dblp: 6, Scopus: 102, Springer Link: 258). Due to the differing search syntax of the respective databases, the search strings had to be adapted and split up so that as many relevant studies as possible could be found.

Subsequently, duplicates were identified and invalid matches removed. As a result, 54 publications were excluded, and 462 were selected for the first iteration. The inclusion and exclusion criteria were applied to the titles, keywords, abstract, and conclusion if necessary. A total of 397 publications were excluded, and 65 were examined in the second iteration. In this final iteration step, the full-text review was carried out. A total of 42 publications were excluded, and 23, as listed in Table 5, were finally investigated to answer the research questions. Fig. 5 shows the distribution of the identified primary studies by year of publication.

A. ASESSMENT OF QUALITY ASSURANCE

By assessing the previously defined quality criteria in the selected primary studies, a systematic classification is made in terms of transparency and significance shown in Table 6.

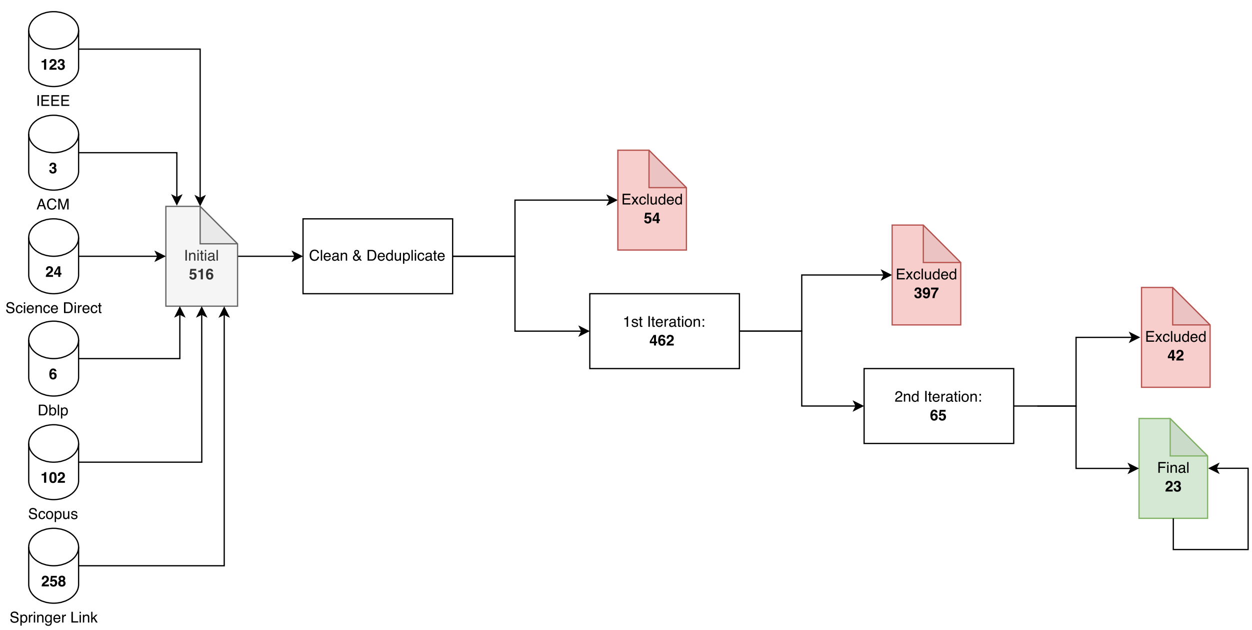

FIGURE 4. Quantitative results conducting phase.

TABLE 5. Finally determined primary publications.

#	Title	Year	Reference
1	OntoSmartDCU: A Multi-Agent Microservice Based Automated, Dynamic Onto-logy and Knowledge Graph Creation Frame-work From Real-Time Sensor Data for the Smart DCU Digital Twin	2024	[41]
2	Multi-agent approach for developing a digital twin of wheat	2020	[42]
3	Hypermedia Multi-Agents, Semantic Web, and Microservices to Enhance Smart Agriculture Digital Twin*	2023	[43]
4	Generation of Asset Administration Shell With Large Language Model Agents: To-ward Semantic Interoperability in Digital Twins in the Context of Industry 4.0	2024	[44]
5	Engineering a Multi-agent Systems Approach for Realizing Collaborative Asset Administration Shells	2022	[45]
6	Dynamic Replanning using Multi-Agent Systems and Asset Administration Shells	2022	[46]
7	A Methodology for Integrating Asset Administration Shells and Multi-agent Systems	2023	[47]
8	Design and Implementation of a Web-based Collaborative Manufacturing System Based on Knowledge Graph	2024	[48]
9	Developing a smart cyber-physical system based on digital twins of plants	2020	[49]
10	Towards the generic integration of agent-based AASs and Physical Assets: a four-layered architecture approach	2021	[50]
11	Towards autonomous system: flexible modular production system enhanced with large language model agents	2023	[51]
12	Machine Learning Agents Augmented by Digital Twinning for Smart Production Scheduling	2023	[52]
13	Cognitive Digital Twins of the natural environment: Framework and application	2025	[53]
14	Agent-Based Asset Administration Shell Approach for Digitizing Industrial Assets	2022	[54]
15	A Quality-Oriented Digital Twin Modelling Method for Manufacturing Processes Based on A Multi-Agent Architecture	2020	[55]
16	Enabling a Multi-Agent System for Resilient Production Flow in Modular Production Systems	2022	[56]
17	An industrial agent-based customizable platform for I4.0 manufacturing systems	2023	[57]
18	On the Use of Asset Administration Shell for Modeling and Deploying Production Scheduling Agents within a Multi-Agent System	2023	[58]
19	An OPC UA model of the skill execution interaction protocol for the active asset administration shell	2021	[59]
20	Production Scheduling Based on a Multi-Agent System and Digital Twin: A Bicycle Industry Case	2024	[60]
21	Multi-Agent Heterogeneous Digital Twin Framework with Dynamic Responsibility Allocation for Complex Task Simulation	2022	[61]
22	MADTwin: a framework for multi-agent digital twin development: smart warehouse case study	2023	[62]
23	The Framework for Designing Autonomous Cyber-Physical Multi-agent Systems for Adaptive Resource Management	2019	[63]

TABLE 6. Results on quality assessment.

	%	#
Q1: Context		
completely fulfilled	100	23
partially fulfilled	0	0
not fulfilled / missing	0	0
Q2: Background		
completely fulfilled	91.3	21
partially fulfilled	4.35	1
not fulfilled / missing	4.35	1
Q3: Outlook		
completely fulfilled	56.52	13
partially fulfilled	34.78	8
not fulfilled / missing	8.7	2

All of the 23 selected studies provide an adequate context for their implementation, mainly due to the thorough full-text review. Technical background information on the methods and technologies applied, for example, in the form of explanatory references and existing approaches, is comprehensively and completely taken into account in 21 publications, while in one case it is too brief, unclear, and in another case it is not specified. However, only 13 of the studies provide an adequate outlook on future research. Eight others take the quality criterion into account with restrictions, and two do not address this aspect at all. None of the studies evaluated were of such poor quality that they had to be excluded from further analysis.

B. FINDINGS

It should be noted that the research approaches of the author groups were considered holistically and that, in some cases, the entire research approach is used to evaluate the review in order to establish a proper analysis subsequently.

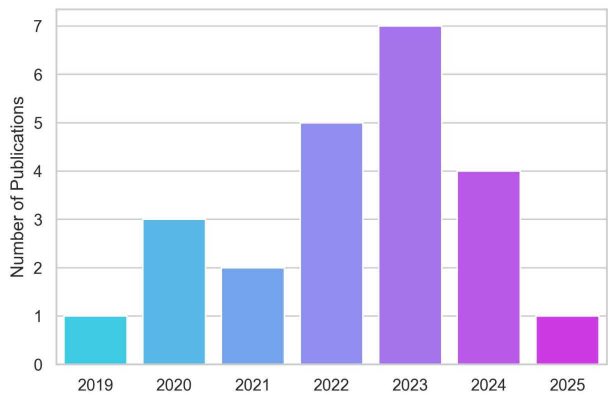

FIGURE 5. Publications per year.

1) FINDINGS RQ 1-APPLICATION DOMAIN

The application domains of the examined studies are shown subsequently. According to Fig. 6, 18 studies deal predominantly with topics that can be assigned to manufacturing and production in the context of Industry 4.0. The remaining studies deal with more specific areas of application, such as 3 publications in smart agriculture and 2 publications in smart cities & environment.

2) FINDINGS RQ 2-DT STRUCTURES AND CONCEPTS

In order to better understand the structure, the necessary frameworks, and concepts of the various DT approaches, the following section examines these in more detail. First, the depth of data- and DT integration was examined in RQ 2.1. For this purpose, the categorization between digital

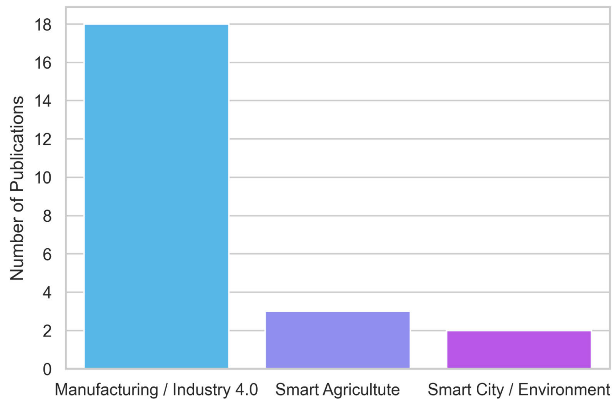

FIGURE 6. Publications per domain.

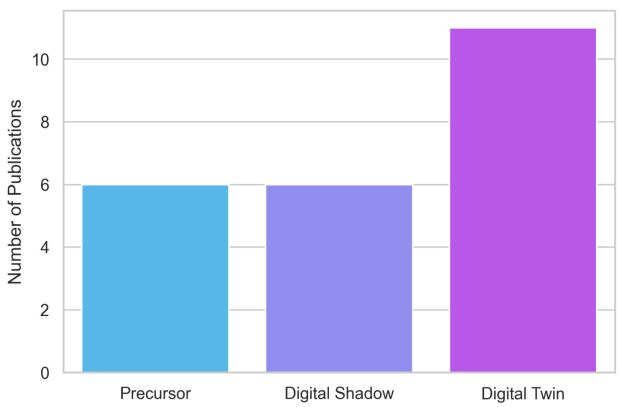

FIGURE 7. Publications per type.

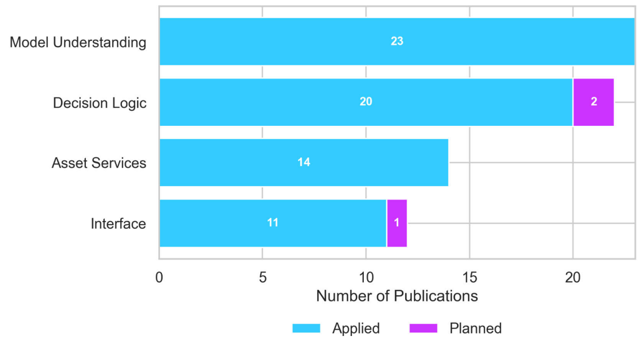

FIGURE 8. System capabilities.

model, digital shadow, and DT was carried out, aligned with the literature outlined before [15], [16]. The analysis shows that not all cases in which the term DT is used actually refer to a fully-fledged DT as per the definition. Taking into account this distinction, the studies can be divided into systems with automatic, bidirectional information flow, i. e., DTs, and digital shadows with no automatic feedback to physical assets. Due to the need for a more representative form for the approaches described in this review than the digital model, no study could be identified under this category. However, some stages are conceptual approaches that perform active data integration only at a descriptive level without a valid digital shadow or DT. These approaches have therefore been classified in the extended category of precursors.

According to Fig. 7, the 6 precursor publications provide fundamental concepts, architectures, and modeling methods without instantiating any real connection. Digital shadows and DT systems can be built based on these as reference models. 6 digital shadow approaches were identified. In [52], for example, the interaction of the DT is limited to the agent ecosystem and does not include any connection to other enterprise systems or physical assets. However, as in most studies involving digital shadows, the possibility of future integration into physical control processes is often mentioned as a prospect.

Among the publications analyzed, 11 explicitly address fully evolved DTs and describe autonomous control commands without human intervention. The agricultural and smart city/environment publications are noteworthy for their higher level of integration of digital shadow and DT. In forecasts, models, and simulations, these suggest recommendations for decisions and control [41], [42], [43], [49]. Despite the differences described, all studies are summarized under the term DT for the following analysis in order to ensure a uniform structure.

Due to the strong representation of manufacturing studies in the review, efforts toward standardization approaches through AAS, driven by the Industry 4.0 initiative, should be highlighted according to RQ 2.2. The majority of studies natively utilize AAS as an interface. It is particularly favored as a basis for MAS used in production for negotiations and intelligent decision-making. The distribution of all identified publications according to their use of AAS shows its utilization within 13 approaches as a framework for DTs. The reasons for employing AAS in the studies are consistent with the previously outlined literature. Specific factors mentioned include compliance with standardization strategies [54], use for data modeling and communication to improve system scalability [60], and standardized access to assets [45]. Some publications refer to the AASX Package Explorer as a graphical user interface for creating the AAS [44], [45], [46], [47], [54]. Eclipse Basyx is mainly employed for the provisioning and operational management of the AAS (e. g., [46], [47]. However, AAS is currently only used in manufacturing publications. Agriculture, Smart City & Environment do not use standardized DT concepts.

In order to take a step toward intelligent DT behavior with RQ 2.3, the studies were examined with regard to the intelligent structural DT features according to [20] as introduced previously. Fig. 8 illustrates the extent to which these features were employed in the studies examined.

The overall model comprehension results from the subfocus on semantic enrichment. Accordingly, semantic technologies or standardized data models are used in all

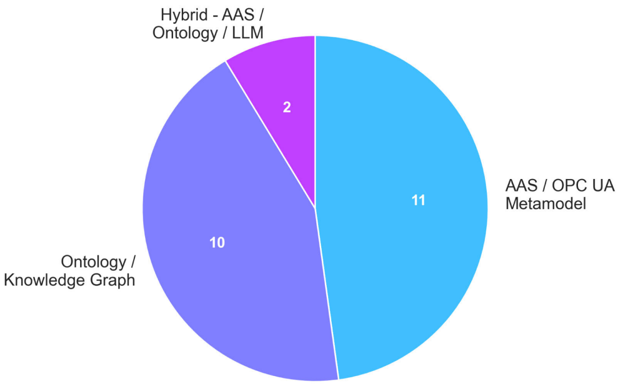

FIGURE 9. Publications per semantic enrichment approach.

23 publications. Approaches to DT decision logic have been implemented in 20 studies and are planned in two further studies. The approaches significantly vary in terms of their specific implementation, for example, based on agent-based decision logic [45], machine learning [52], and Artificial Intelligence (AI) algorithms [54]. [47] specifies proactive AAS as a decision-making unit that can be supplemented by MAS with distributed intelligence. Intelligent decision-making processes are thus addressed extensively, but their functional depth is highly dependent on the respective application context. The aspect of services, i. e., the functions of the physical asset represented in the DT, is addressed in 14 studies. In the available publications, submodels of the AAS are extended to include the skills of the asset, allowing for their representation in a standardized and structured manner [46]. This provides machine functions holistically and makes them dynamically usable for others. Finally, in 11 studies, a specific interface for controlling physical assets is available, and in another 2, it is planned.

3) FINDINGS RQ 3-SEMANTIC ENRICHMENT

After the characteristics and conventions of DT have been identified in the context of the publications examined, the next step is to focus on the specific semantic technologies utilized. Therefore, the studies are first examined based on the conceptual approach of semantic enrichment according to RQ 3.1.

Regarding Fig. 9, the results could be classified into two main and one hybrid category. The approaches based on the inherent AAS or Open Platform Communications Unified Architecture (OPC UA) metamodels were the most frequently represented, with 11 publications. Another 10 publications used ontologies and knowledge graphs to represent semantic system relationships, while two used a combination of both and further employed Large Language Models (LLMs).

Unstructured measurement values, for instance, are semantically annotated during the runtime using specific existing domain ontologies [41]. Particularly in agriculture, a subsequently necessary semantic understanding is created using plant growth ontologies [42], [49]. However, most production-related publications use the inherent metamodels of AAS or OPC UA. Unlike these approaches, LLMs can be utilized to classify unstructured knowledge, such as manuals or humanoid inputs, in order to further enrich an existing knowledge base [44].

Furthermore, RQ 3.2 examines technologies and standards used for the semantic approaches of the individual studies. As explained above, two main groups of semantic enrichments are formed. These groups also use different fields of technology for this purpose. AAS-oriented approaches use their own meta model, which can be exchanged via JavaScript Object Notation (JSON) or Extensible Markup Language (XML) formats, for instance, or in reactive applications via AAS servers and registries, for example, using Representational State Transfer (REST) [44], [57], [58]. In this context, ECLASS can also be utilized as a semantic alignment for ground truth [44], [56]. OPC UA [59] and AutomationML [63] can also serve as a metamodel for a semantic background.

The other group of ontology and knowledge graph-based approaches uses the standard technology stack of semantic web technologies, such as Resource Description Framework (RDF) and Web Ontology Language (OWL) [41], [55], [61], [63], as query languages SPARQL Protocol and RDF Query Language (SPARQL) with Semantic Web Rule Language (SWRL) or Cypher [63] in widely used graph databases, such as Neo4j [63]. Ontology engineering often is accomplished using Protégé [55], [62].

4) FINDINGS RQ 4-MULTI AGENT SYSTEMS

Once the basis of semantic comprehension has been examined, it is necessary to focus on approaches that enable proactive behavior through MAS. Therefore, their role and integration within DTs should be examined in more detail through RQ 4.1. As previously introduced, the methodology according to Pretel et al. [37] will be followed. Originally, a distinction is made between DT-oriented (MAS with DT) and DT-guided MAS (MAS for DT). In DT-guided MAS, the MAS serves as the architectural basis and the active component for developing the DT. In DT-oriented MAS, on the other hand, the MAS takes over the functionalities of the DT, such as simulation and real-time data acquisition. In addition, a primary DT-oriented MAS is proposed as a supplementary category to cover the frequently occurring situation in which the MAS functions primarily as a component of the DT architecture, but at the same time accesses the DT, for example, as a data and service source, in order to execute its tasks.

Figure 10 shows the resulting distribution of the studies examined. A total of 20 studies could be assigned to the group of DT-oriented MAS, with a further 3 classified as DT-guided MAS. In addition to the implementation and expansion of the MAS, additional specific features could be identified. Thus, agent-based approaches to active semantic enrichment were used in three studies [41], [43], [44]. In

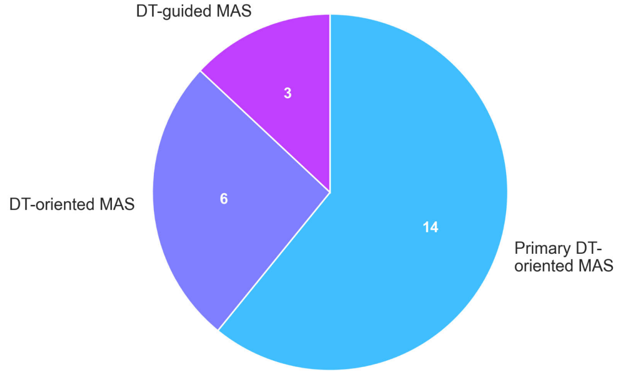

FIGURE 10. Publications per integration depth DT / MAS.

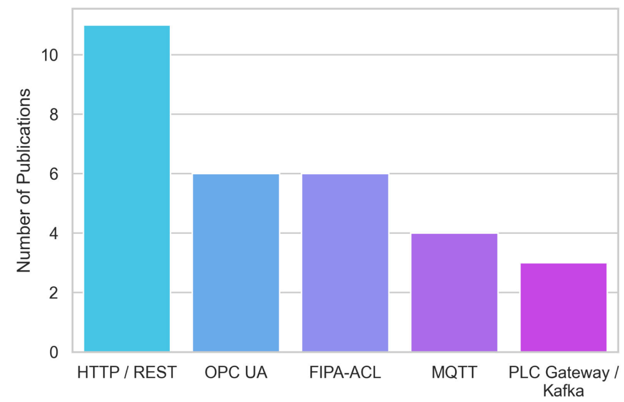

FIGURE 11. Publications per communication protocols and standards of DT / MAS / asset.

13 studies, sophisticated distributed decision-making takes place through the MAS [42], [45], [46], [47], [48], [50], [51], [52], [53], [54], [55], [57], [62].

The design of the communication structure within the approaches, especially between MAS and DT, is examined in RQ 4.2. Three different types of interaction could be classified in the respective communication hierarchies, each with typical protocols/standards: Agent-DT communication, whereby agents communicate with the respective DTs of the machines, Agent-Asset communication, in which agents interact directly with the machines they represent, and Inter-Agent communication, in which different agents communicate with each other bidirectionally. In summary, this results in the distributions shown in Fig. 11, but in larger numbers than publications, as it was possible to use multiple protocols and standards per study. Other domain-specific communication protocols, such as MTConnect for machine tools, are also relevant in practice but were not covered in the primary studies examined.

It is noticeable that the technology stacks of the studies that use AAS often employ Hypertext Transfer Protocol (HTTP)/REST, Message Queuing Telemetry Transport (MQTT), and OPC UA in combination for Agent-DT interaction. OPC UA offers direct asset interaction through executable machine skills via OPC UA methods, events, and state machines [45], [47], [58].

Apache Kafka and OPC UA are suitable for field communication between agents and assets for asynchronous dissemination and coupling [45], [47], [54], [58], [59], [63]. PLC gateways or MQTT approaches are used as special cases for data exchange directly between agent and machine [50], [57], [62].

FIPA-ACL, according to IEEE 2660.1-2020 [64] is the approach of choice for inter-agent communication in established MAS systems [45], [47], [50], [57], [58], [62]. In some cases, the studies use the I4.0 language according to VDI/VDE2193 [65] or via communication middleware [46], [56], [59].

The behavior and interaction of DTs and agents can be modeled using different methods. These methods describe how agents make decisions to achieve their goals. RQ 4.3 explains in more detail which specific implementation approaches were used in the examined studies. 8 studies used established MAS tools to build a proactive orchestration and negotiation structure, while the majority, with 15 publications, relied on self-developed, custom-built frameworks. The existing frameworks primarily include JAVA Agent DEvelopment Framework (JADE) and Janus Agent and Holonic Platform based on the SARL framework. Both frameworks offer the advantage of creating a standardized, semantically oriented language layer between agents for their orchestration and communication, either directly or based on FIPA-ACL [45], [46], [47], [50], [56], [57], [58], [62], [63]. Consistent modeling and interfacing of machine skills and their negotiation play a central role in all approaches. In a direct comparison, JADE is primarily used due to its broad FIPA standard support, integrated tools, and modular structure, while Janus SARL offers particularly flexible customization options for individual capabilities. Concerning the negotiation process and contracting, publications that attempt to implement proactive AAS using I4.0 language, which lays the foundation based on FIPA, stand out [46], [56], [59].

However, the custom-developed frameworks are less flexible in terms of negotiation. Microservice-based agent structures can be used for this purpose [41], [43]. The focus is on event-based sequences, collaboration, state-machines, and rescheduling algorithms for process optimization [43], [49], [53], [60], [61], or further the prediction of states [42]. LLMs are also used to contextualize agents via structured prompts, thereby integrating them into the system [51].

Interaction patterns, which range from formal negotiation protocols to lightweight service orchestration, are examined in RQ 4.4. With regard to negotiations, several FIPA-applying studies primarily implement contract networks and auction families to distribute tasks among resources [45], [56], [58], [63]. Industry 4.0-anchored approaches leverage active AAS and the I4.0 language, coupling state machines of requesters and providers preceded by a bid and feasibility phase when required for execution [46], [56], [59]. Apart from explicit negotiations, collaboration is often achieved through ontology/knowledge graph-mediated orchestration or event-driven coordination, which schedules services without market-like behavior [44], [53], [55], [61]. Overall, the prevailing pattern in industrial DT and MAS contexts is a request-response with semantic grounding via AAS, metamodels, or ontologies, whereby formal negotiations are selectively applied for resource allocation.

5) FINDINGS RQ 5-LIMITATIONS AND FUTURE WORK

As a final focus, the limitations and future work are consolidated under RQ 5. The majority of studies claim generic limitations and future research directions. These include expanding use cases, conducting extensive evaluations through benchmarking, and advancing standardization initiatives.

These standardization gaps and efforts, particularly with regard to AAS approaches, are explicitly addressed in terms of AAS submodels, the integration of production and interoperability standards [45], [60], [62]. Dynamic negotiation behavior in collaboration, scheduling, and rescheduling must be further refined through additional cognitive capabilities under changing conditions with extended production plans [41], [46], [54]. Another key challenge lies in retrofitting existing systems, the design of services for comprehensive diagnosis, and the scope for systematic benchmarking [51], [54]. An additional benefit would arise if the frameworks were expanded to other industrial sectors through the adaptation of agents and information models [60].

V. DISCUSSION

The following section synthesizes the results of current research efforts and outlines future trends. The 23 primary studies assessed in the full-text review covered various areas of application, ranging from smart agriculture to smart cities, but mainly dominated by production and manufacturing contexts. Driven by the Industry 4.0 initiative, intelligent DTs remain significant value drivers and offer opportunities for dynamic scheduling, orchestration, and reconfigurability. Semantically enriched MAS-DT approaches create a viable technical path toward intelligent negotiation behavior of DTs. Proof-of-concept adaptations in other application domains demonstrate the transferability of the underlying design principles and the potential for cross-domain use beyond production environments.

The integration depth with regard to the bidirectional integration between DTs and physical layers exhibits mixed approaches. While a substantial subset of the studies already operates autonomously with minimal manual intervention, the other subset operates at a digital shadow level without closed-loop actuation, inherently hindering independent agent-based interaction and constraining truly proactive behavior. The precursory approaches and the overall limited number of end-to-end implementations suggest that the combined MAS-DT semantic stack is still in an immature stage of consolidation.

A widely adopted approach in manufacturing studies is standardization through AAS, which in turn ensures continuous connection to the physical level in a middleware configuration through defined interfaces and data structures. This advantage is neglected in other application domains, resulting in many custom frameworks with limited standardization, interoperability, and scattered information models. However, the AAS is limited in terms of decision logic and intelligence and requires additional software components and MAS approaches to be implemented. While all studies assume a semantic model understanding and, for the most part, also take steps toward decision logic, the production approaches distinguish themselves with direct access and interaction with executable machine skills. In this context, the paradigms of skill-based approaches and their semantic representation and interaction are dominant as a basis for agent-based negotiation of a production-optimum. By adapting it accordingly, this would also apply to all other domains.

Semantic representation, which is a prerequisite for all approaches, can be classified into two general categories. Half of the studies utilize the metamodel of AAS as a semantic backbone, while the other half relies on explicit knowledge modeling, as exemplified by ontologies and knowledge graphs to formalize entities, relations, and constraints. A small subset of the studies utilizes LLMs to interpret unstructured data formats in order to make them usable in a structured form. These emerging approaches offer promising prospects for reducing manual setup effort through LLMs, thereby simplifying the provision of DTs, but still at an early stage of development. LLMs can be positioned as complementary tools to formal semantic models for extracting entities and relationships from unstructured documents, assisting with metadata tagging and semantic annotation, and supporting mapping tasks between heterogeneous objects and interface descriptions. Nevertheless, LLMs should not be considered a substitute for formal semantics when deterministic behavior, verifiability, and safety guarantees are required, as current LLM-based pipelines generate non-deterministic results, and it remains difficult to validate and certify them for safety-critical industrial control systems, in addition to latency and resource constraints in real-time environments.

MAS patterns can be divided into various integration depths and the extent to which they enable decision logic and behavior. A large majority of studies go beyond pure information retrieval from the DT and can be characterized as DT-oriented MAS, where agents assume subtasks of intelligent DTs such as simulation, coordination, and negotiation. The employed implementations also diverge between established MAS frameworks, such as JADE or Janus SARL, in coexistence with custom-built systems. Communication stacks typically combine common IoT protocols such as MQTT, HTTP, and OPC UA, while FIPA-Agent Communication Language (ACL) is often utilized for formalized inter-agent communication. Particularly in a few advanced heterogeneous manufacturing approaches, Industry 4.0 language is established, enabling future direction toward integrated proactive AAS to be leveraged towards independent negotiation. In contrast, other approaches query ontologies or knowledge graphs implementing independent interaction patterns without explicitly formalized negotiation protocols.

Concluding, intelligent behavior in DTs can be described as the interplay of semantic modeling, reasoning, and agent-based decision-making. Semantic modeling provides a machine-readable representation of the state, context, and capabilities of the DT (e.g., via information models, knowledge graphs, or ontologies). Reasoning processes derive decision-relevant abstractions from this semantic level, such as goals, constraints, and possible actions. Agent-based decision-making then implements these abstractions through decentralized coordination, negotiation, and action selection. A closed-loop behavior emerges when selected actions are bound to executable skills at the physical level, and the execution results are fed back to update the state of the DT and its semantic representation, enabling continuous re-evaluation and adaptation. As a possible operationalization anchor, the Digital Twin Reference Model (DTRM) [66] can serve as an implementation-oriented conceptual design to make these interactions explicit and link the SLR results to a coherent architectural design paradigm.

Across the corpus of primary studies, the systematic investigation of limitations and threats to validity should be noted. Most studies are prototypical, addressing partial aspects without comprehensive and methodologically sound validation, while future work is mostly superficial and general in scope. This results in a strong focus on architectural prototypes rather than frameworks with comparable performance. In practice, recurring implementation barriers hinder end-to-end realization. These include, in particular, integration into brownfield systems with heterogeneous interfaces and data structures, scalability limits when increasing the number of assets, and the lack of controllable semantic mappings. Furthermore, real-time capability and determinism in a production environment remain critical development factors, including agent lifecycle management, monitoring, and error handling. These factors account for fewer reports on closed-loop actuation than on advisory or monitoring decision support tools. Future research should prioritize measurability and comparability, as well as unification and standardization, in order to make DTs more intelligent and proactive across application domains.

In summary, the manufacturing sector, driven by Industry 4.0 goals, is emerging as the dominant driving force behind proactive intelligent DTs. With the key consensus technologies of AAS already established, there is an information and data basis, knowledge graphs for semantic abstraction, skill-based control structures, and decentralized decision-making, as well as negotiation logic through MAS, and integrated architectures are already being applied in research. The review has shown that other sectors have already addressed similar issues and that the transfer of technologies used in manufacturing can also serve as an adaptation aid and become a catalyst for change in other domains. The most significant general gaps identified are consistent evaluation and benchmarking, semantic interoperability across domains, and the establishment of standards as well as frameworks.

VI. CONCLUSION

This literature review synthesizes current research findings on intelligent DTs by means of merging semantic technologies and MAS. The insights reveal a fragmented development process that is still in its beginnings. While the range of applications extends from manufacturing to smart agriculture, to smart city and environmental settings, current innovation is strongly driven by production-oriented scenarios based on Industry 4.0 paradigms.

Nevertheless, DTs are still deployed inconsistently. Many of the approaches are prototypical or are rather vague, lacking bidirectional, autonomous integration between the physical and digital spaces. This discrepancy determines whether agents can move beyond advisory optimization to closed-loop control and proactive behavior. Semantic comprehension, as a necessary prerequisite for coordination and decision synthesis, can be divided into two approaches: AAS metamodel-centered approaches with standardized interfaces specifically for production, and ontology or knowledge graph-centered approaches that enable explicit formalization of entities, relationships, and constraints. Hybrid approaches begin with the integration of LLMs to organize unstructured data, but in the literature, these are positioned more as supplementary than as a replacement for formal semantics. The realization of autonomy via MAS is integrated with a predominant DT orientation, whereby agents take on important DT functions such as simulation, coordination, and negotiation, rather than merely accessing their state representation. Manufacturing also proves to be dominant here, making use of established frameworks and communication standards for advanced production configurations.

Overall, the studies examined confirm that semantically enriched MAS-DT architectures represent a viable technological path to proactive, intelligent DT negotiation behavior. However, domains beyond manufacturing are still in an early consolidation phase, characterized by simple, broken-down implementations and limited comparability between studies.

Future research must shift from prototypical demonstrations to comparable and transferable frameworks by establishing unified benchmarks, strengthening cross-domain semantic interoperability, and systematically evolving implementations to closed-loop actuation with adequate governance and runtime security. In addition, challenges arise in the brownfield integration of such approaches, as well as important leveraging technologies such as LLMs, in order to ensure trustworthiness and, where necessary, determinism. The multitude of pioneering approaches from manufacturing must be transferred to other domains to benefit sustainably from synergy effects. It is also important to ensure that research results can be transferred into robust, scalable industrial applications in order to popularize the feasibility and benefits.

ACKNOWLEDGMENT

During the preparation of this work, the authors used ChatGPT, DeepL Write, and Grammarly AI-based language models to improve language and readability. After using these tools, they reviewed and edited the content as needed and take full responsibility for the content of the publication.

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